EES301: Global biogeochemical cycles and Climate change

Biogeochemical cycling of major elements

EE301
Dr. Sunil A. Patil

- Global element distribution and environmental processes
- Movement and transformation of elements through Earth's reservoirs
- Influence of geological, biological, and anthropogenic factors
- Operate on various timescales from days to millions of years

Why is it important to study biogeochemical elemental cycles?

- Utmost importance for the sustenance and balance of ecosystems and the provision of ecosystem services
- Regulate the availability and distribution of essential elements required for life
- Have significant impacts on climate and weather patterns
- Help understand how ecosystems respond to anthropogenic activities
- Contribute to the regulation of nutrient cycles in agricultural systems, impacting food production and global food security

Introduction

- Cycling of elements on Earth involves

  - Geological and geochemical processes, such as erosion, water drainage/runoff, movement of the continental plates & weathering…

  - Chemical and biological processes

  - Interactions between biotic and abiotic components in environments

- Chemical elements move through both biotic (i.e. biosphere) and abiotic compartments (atmosphere, hydrosphere, and lithosphere) on Earth - Inputs, outputs, sources and sinks

- Elements (such as C, N, P and S) take a variety of chemical forms, and are cycled and reused within and among Earth's various compartments over and over again – (natural) **Biogeochemical cycles.**

- Along with energy flows, biogeochemical cycles establish the relations among ecosystem compartments (biotic and abiotic) at local, regional and global scales.

- The total biomass, metabolic diversity, and presence in almost all habitats on earth signify the crucial and dominating roles of **microbes** in the cycling of different elements.

- All living beings play some role - Prokaryotes play the major and sometimes unique roles because of their diverse metabolic capabilities.

The Global Carbon Cycle

- Carbon reservoirs on Earth
- Carbon cycling within and between reservoirs - Key processes
- Anthropogenic effects on the global carbon cycle/budget

CO_2, a fully oxidized version of carbon serves as the basic building block

Major ‘C’ reservoirs

Major Carbon Reservoirs on Earth

Reservoir	Percent of Total[a]
Rocks and sediments	99.5[b]
Oceans	0.05
Methane hydrates	0.014
Fossil fuels	0.006
Terrestrial biosphere	0.003
Aquatic biosphere	0.000002

[a]Total carbon, 76×10^{15} tons
[b]80% inorganic

Brock Biology of Microorganisms, 15th edition

Pools	Quantity (Gt)
Atmosphere (CO_2, CH_4, CO)	720
Oceans	38,400
Total inorganic	37,400
Surface layer	670
Deep layer	36,730
Total organic	1,000
Lithosphere	
Sedimentary carbonates	>60,000,000
Kerogens	15,000,000
Terrestrial biosphere (total)	2,000
Living biomass	600–1,000
Dead biomass	1,200
Aquatic biosphere	1–2
Fossil fuels	4,130
Coal	3,510
Oil	230
Gas	140
Other (peat)	250

Falkowski et al. 2000, Science

- The balance between carbon release from lithosphere and sedimentation rate of carbon in deep oceans (i.e., geological processes) determines the long-term CO_2 level in the atmosphere.

- The amounts of elemental carbon (e.g., graphite and diamond) are insignificant in the carbon cycle.

- **Oxidation states: +4 to -4**

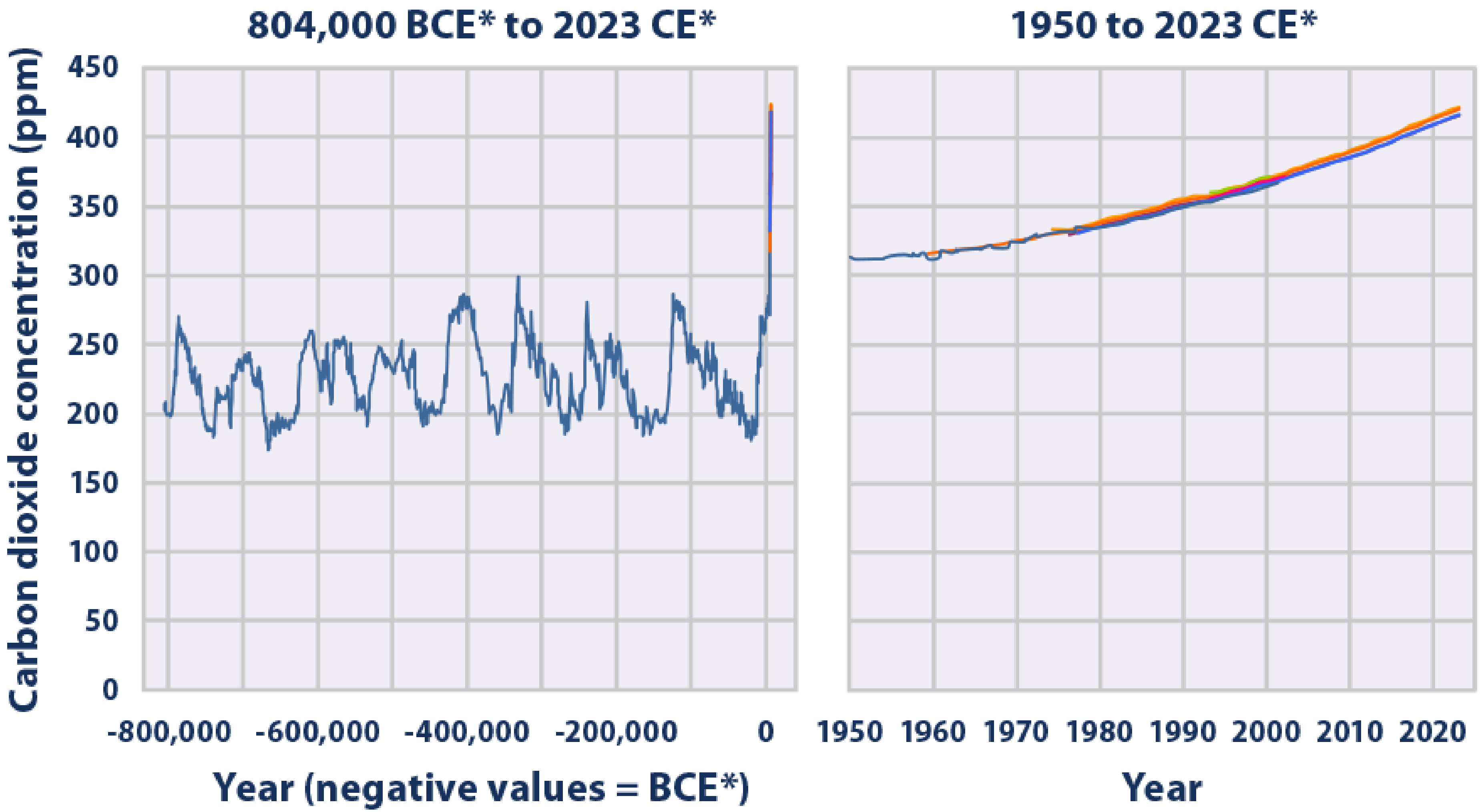
  CO_2: +4 Carbonates: +4 (Oceans and Lithosphere)

  CO: +2 CH_4: -4 Reduced carbon: variable

- **Isotopes of carbon: ^{10}C to ^{16}C (#7)**

  Stable: ^{12}C (99%) and ^{13}C (1%) and others radioactive -^{14}C Radiocarbon

Atmospheric CO_2 concentration



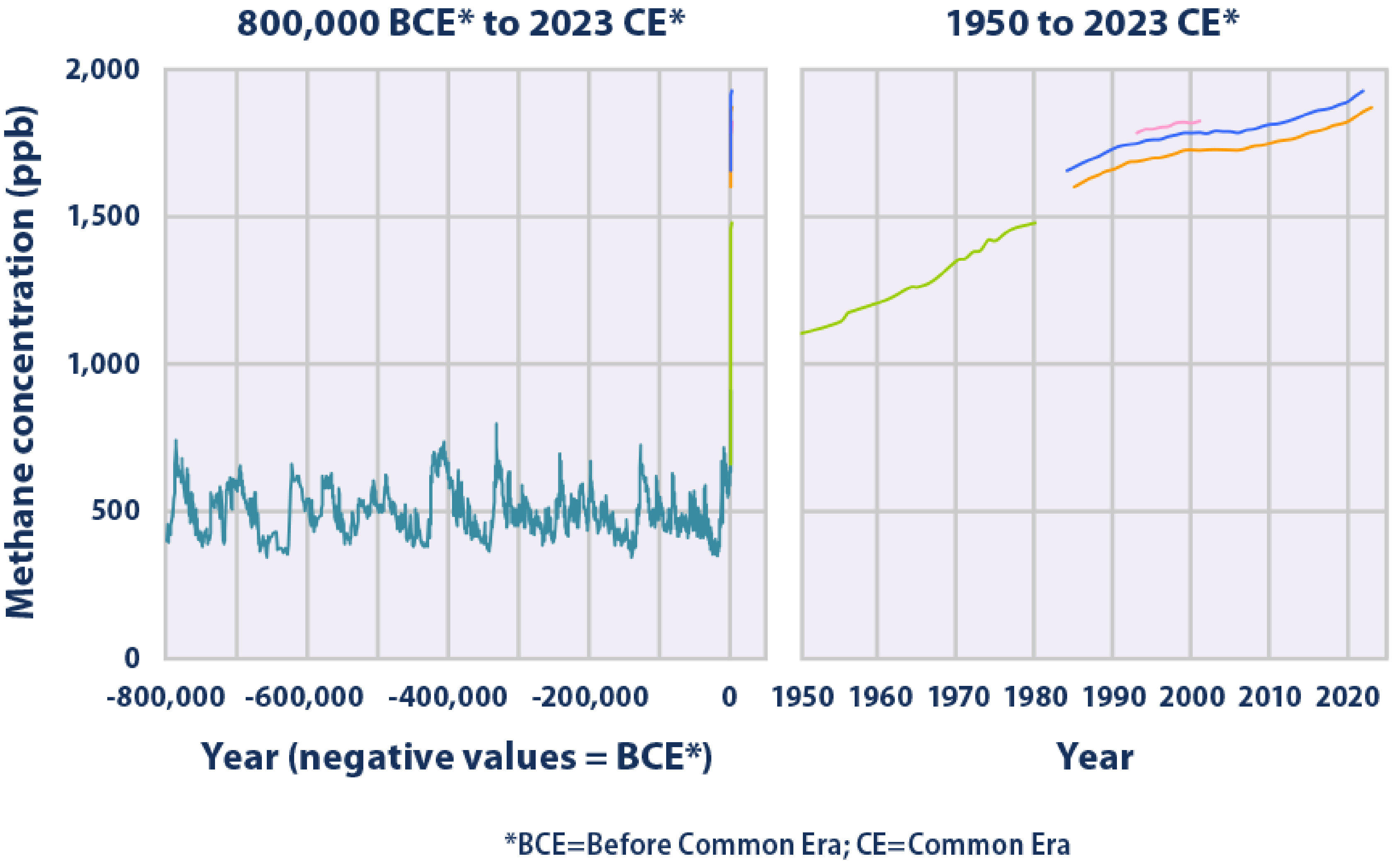
CO_2 rise from an annual average of 280 ppm in the late 1700s to >422 ppm in 2024 — a >50 % increase.

~1.8 ppmV/yr (4 PgC/yr) increase rate

*BCE=Before Common Era; CE=Common Era

https://www.epa.gov/climate-indicators

Methane – about 1% of the atmospheric carbon budget (~3 Pg C)



CH_4 oxidation is one of the source of CO in the atmosphere – 0.05 to 0.20 ppmv (~ 0.2 Pg C)

Other forms (e.g., terpenes, hydrocarbons, dimethyl sulfide, etc. – 0.05 Pg C

https://www.epa.gov/climate-indicators

Oceanic carbon

- Dissolved inorganic, dissolved organic, particulate organic, fixed C (marine biota ~3 Pg C)

- Governed by the Carbonate chemistry

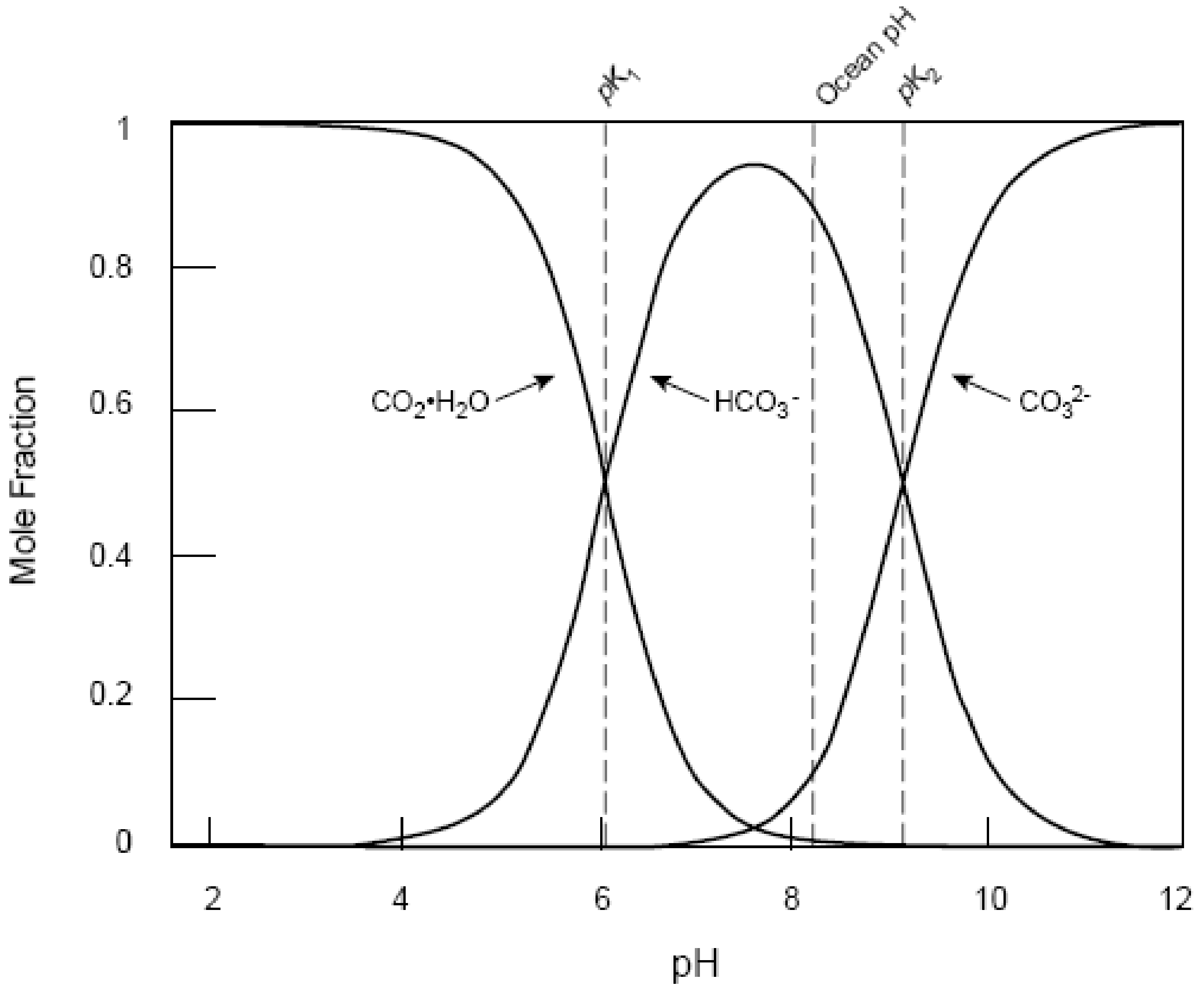
$$CO_2(g) \overset{H_2O}{\Leftrightarrow} CO_2 \cdot H_2O$$

$$CO_2 \cdot H_2O \Leftrightarrow HCO_3^- + H^+$$ **pKa = 6.1**

$$HCO_3^- \Leftrightarrow CO_3^{2-} + H^+$$ **pKa = 9.2**

- Average Ocean pH = 8.2; maintained by weathering of basic rocks

Speciation of total carbonate, CO_2 (aq) in seawater Vs pH



Since pKa1< pH < pKa2 , most of the dissolved CO_2 exists as HCO_3^-

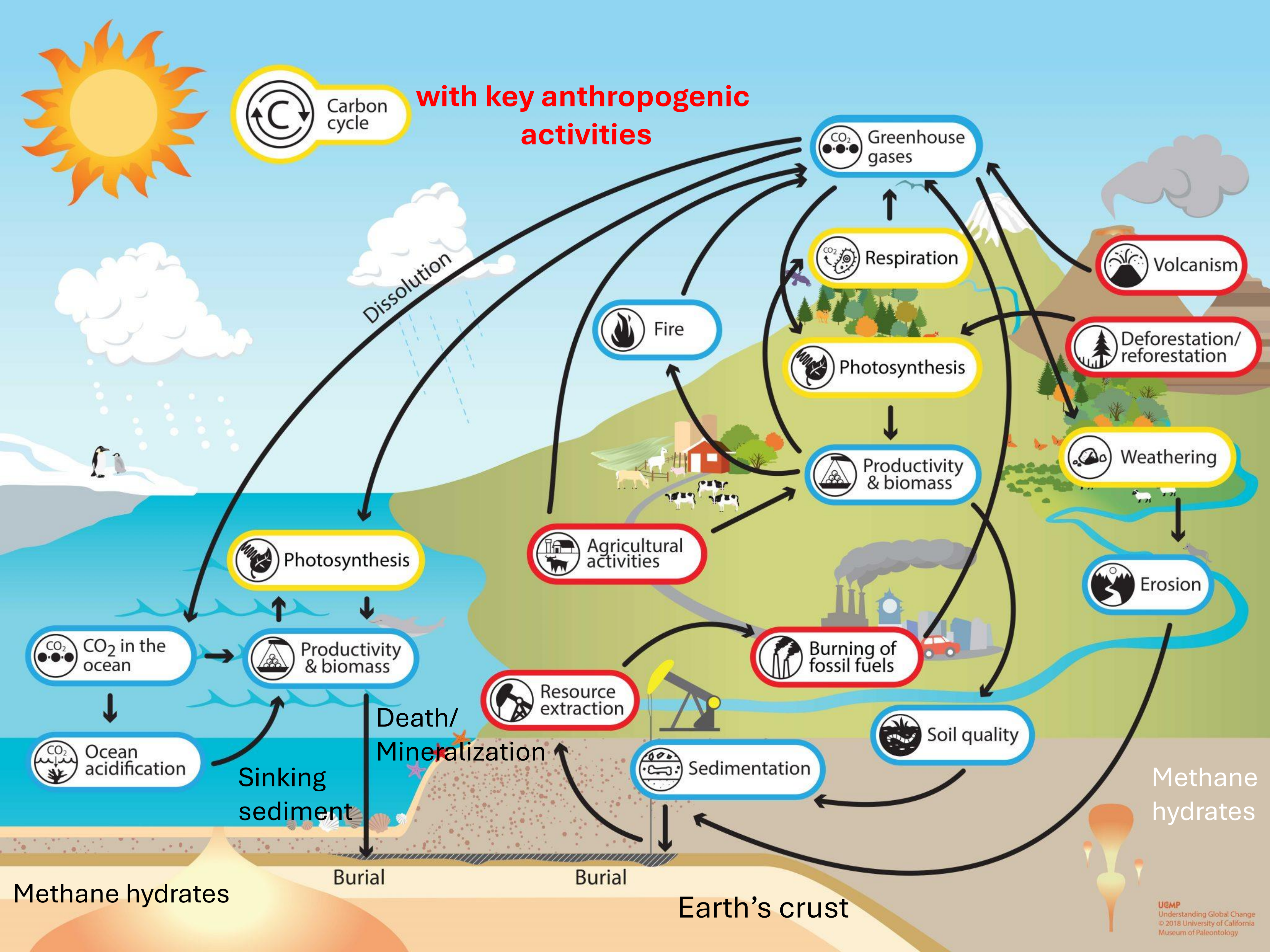
Ocean acidification

- from 8.19 to about 8.05

Note: If oceans were acidic, almost all CO_2 would partition to the atmosphere.

Global Carbon cycle

Spanning the atmosphere, biosphere, hydrosphere and geo/lithosphere



1 Gigatonne (Gt) = 1 billion tonnes = 1×10^{15}g = 1 Petagram (Pg)

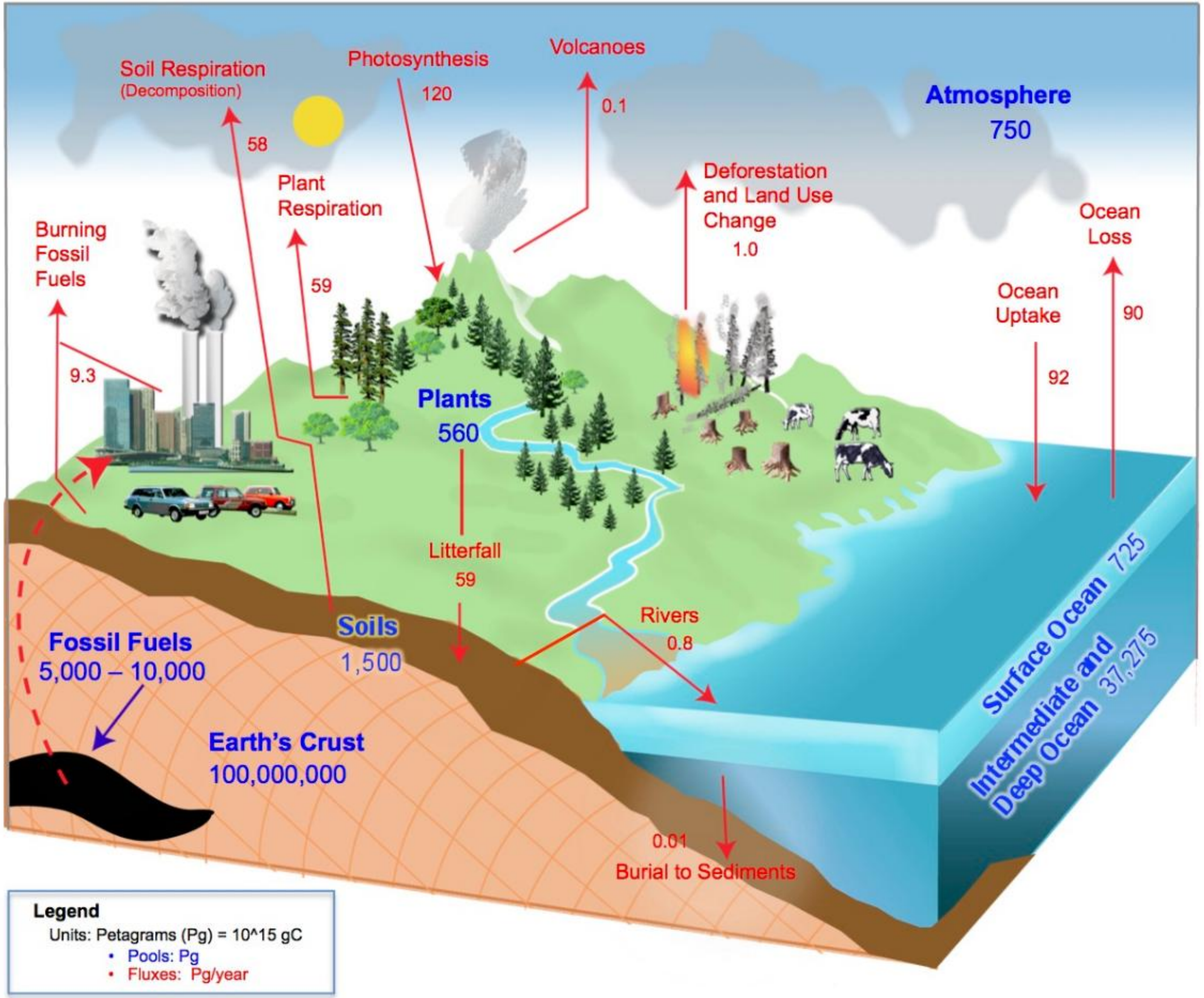
1 kg carbon (C) = 3.664 kg carbon dioxide (CO_2)

1 GtC = 3.664 billion tonnes CO_2 = 3.664 $GtCO_2$

Global Carbon Budget:
Natural and anthropogenic fluxes

Methane flux to atmosphere ~0.5 Pg C/yr

1 Pg C = 1 Gt C



GLOBE®2017 Global Carbon Cycle Diagram Biosphere

Data Sources: Adapted from Houghton, R.A. Balancing the Global Carbon Budget. Annu. Rev. Earth Planet. Sci. 007.35:313-347, updated emissions values are from the Global Carbon Project: Carbon Budget 2017. Diagram created by a collaboration between UNH, Charles University and the GLOBE Program.

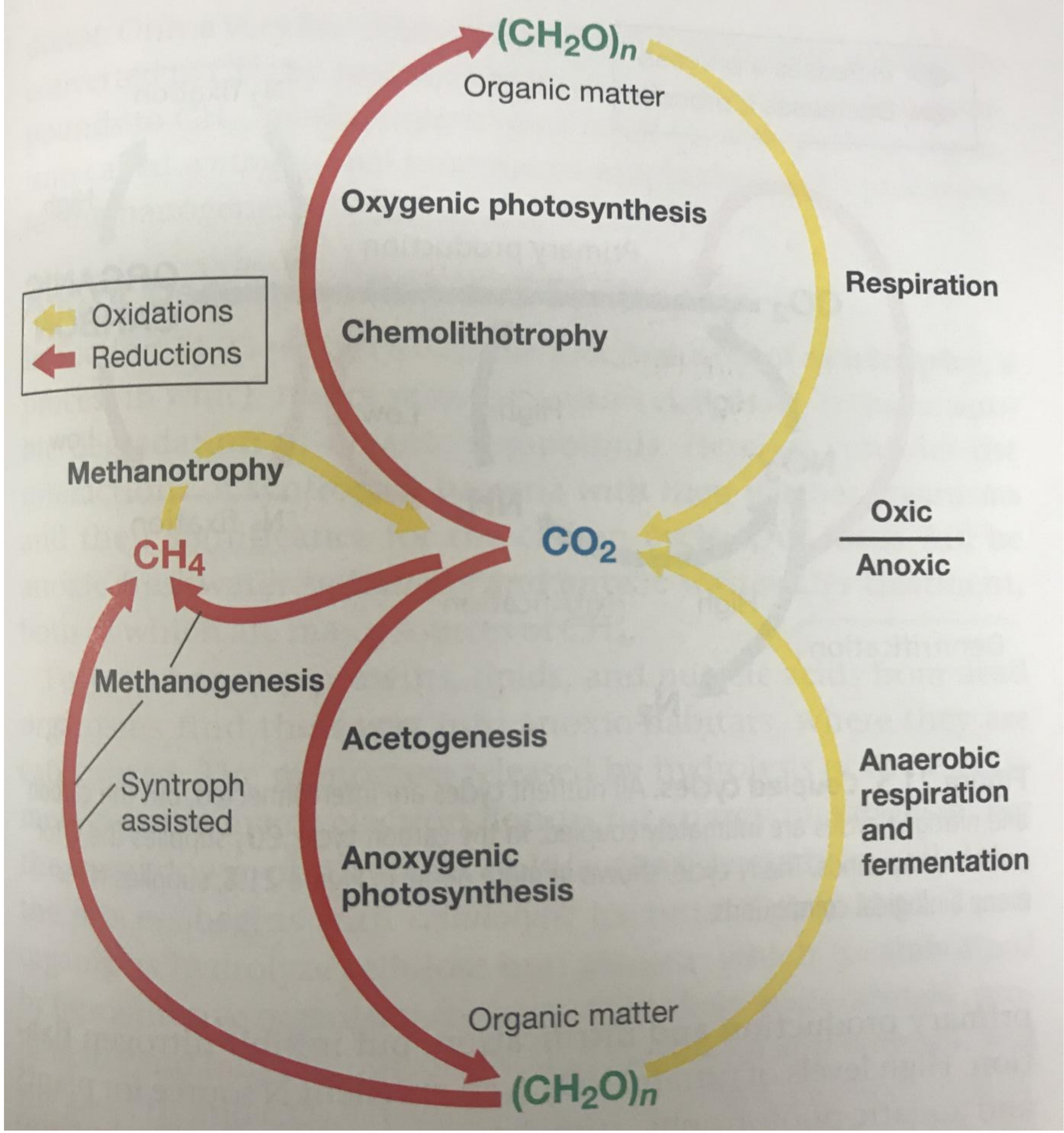
Key processes

- Sedimentation, volcanism, uplifting, erosion/run-off, etc. (Geological)

- Ocean exchange, Dissolution, weathering, etc. (Geo/Chemical)

- **Photosynthesis, Respiration, Decomposition, etc**. (Biological) – major role in cycling of carbon between "fast" reservoirs

- Fossil fuel burning, Land use changes (Agriculture activities, Forestation/ Deforestation, etc.) (Anthropogenic)



Redox cycle for carbon

It contrasts
Autotrophic
(CO_2 ----> organic compounds)
and
Heterotrophic
(organic compounds --> CO_2)
processes

$(CH_2O)_n$ represents organic matter

Brock Biology of Microorganisms, 15th edition

Relationship between autotrophs and heterotrophs

Autotrophy

$nCO_2 + nH_2O$ -------------------> $(CH_2O)_n$

Heterotrophy (respiration)

$(CH_2O)_n + nO_2$ -------------------> $nCO_2 + nH_2O$

- For organic matter to accumulate, the rate of autotrophy must exceed the rate of respiration.

- **Since CO_2 is the most prevalent greenhouse gas in the atmosphere, it isn't good if these two equations get out of balance.**

Photoautotrophy

- Foundation of the Carbon cycle
- Abundant only in habitats where light is available

Oxygenic photosynthesis (major process)

$nCO_2 + nH_2O \rightarrow (CH_2O)_n + nO_2$

H_2O is a source of electrons

Calvin cycle

E.g., Plants, Algae and cyanobacteria (mostly aquatic environments)

Relation with O_2 cycle

Anoxygenic photosynthesis

$nCO_2 + nSH^- \rightarrow (CH_2O)_n + nS + nH_2O$

SH^- or H_2 or thiosulfate or Fe^{2+} is a source of electrons

E.g., Sulfur and non-sulfur bacteria

Reverse Krebs cycle, 3-hydroxypropionate pathway

Relation with “S” and “Fe” cycles

Chemolithoautotrophy

- Dark carbon fixation

- **Reductive/Reverse TCA cycle:** Hydrogen-oxidizing bacteria/Knallgas bacteria, Sulfur-oxidizing bacteria, Iron-oxidising bacteria, Nitrifying bacteria

- Wood-Ljungdahl pathway/ Reductive acetyl-CoA pathway

  Acetogenesis

  $2CO_2 + 4H_2 \rightarrow CH_3COOH + 2H_2O$

- **Methanogenesis**

  $CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$

Organic matter Decomposition + Respiration

- **Release of CO_2 and CH_4**
- Decomposition of organic material under oxic conditions

$$(CH_2O)_n + nO_2 \text{------------------>} nCO_2 + nH_2O$$

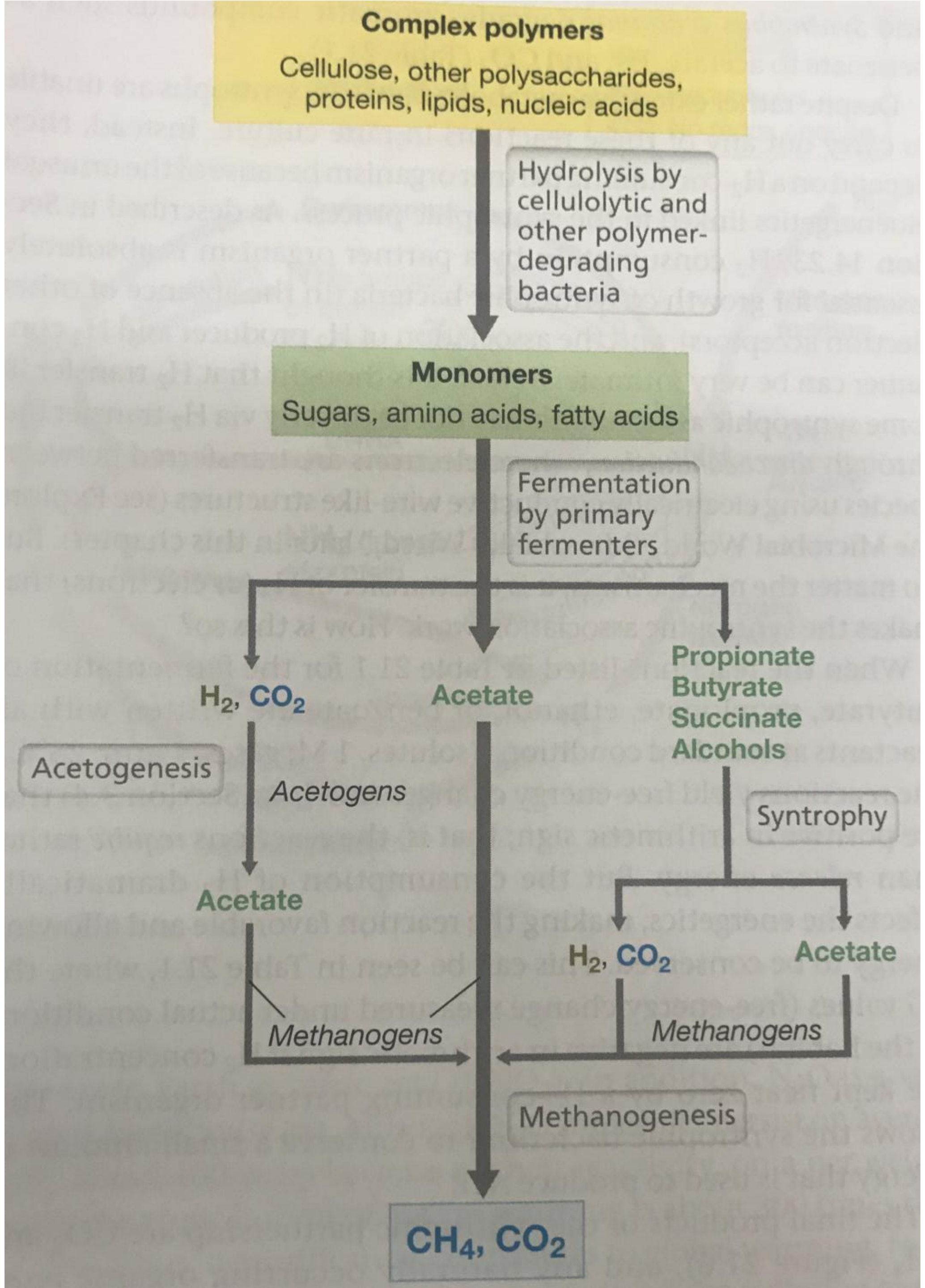
<u>Aerobic respiration</u>

<u>Typical process:</u> initial degradation of polymers - depolymerization

polymers (e.g. cellulose) ------------------> monomers (e.g. glucose)

nGlucose + nO_2 ------------------> nCO_2 + nH_2O

- <u>Anaerobic respiration</u>: other electron acceptors (e.g., nitrate, sulfate, Fe^{3+})



Anoxic decomposition

common in anoxic environments/sub-surface sediments, water-logged soils, wetlands, wastewater treatment processes, rice cultivation, and ruminants

Brock Biology of Microorganisms, 15th edition

Where Methane Meets Microbiology

THE MICROBIAL ROLE IN CLIMATE CHANGE

Methane is the largest component of natural gas.

It is the second most abundant greenhouse gas and the biggest contributor to global warming after carbon dioxide.

Looking at a 100-year horizon, methane has **28 times** more global warming potential (GWP_{100}) than carbon dioxide.

In the past 150 years, methane has added **0.5°C to global warming**.

70% of all methane comes from **microbes**.

This means a large portion of global warming is believed to be microbially controlled.

Humans are responsible for much of these emissions by creating environments where these microbes thrive.

Methanogenesis

FUN FACT
Methanogenesis is estimated to be more than 3 billion years old, making it an ancient metabolism.

Methane is formed as a byproduct when methanogens metabolize their food sources to gain energy. This microbial process of creating methane is called **methanogenesis**.

Methanogenesis can be accomplished by different reactions. Two very common ones are:

**These are two examples of many other kinds of methanogenic reactions using different starting chemicals.*

Methanotrophy

Methanotrophs consume methane to harness energy in a process called **methanotrophy**.

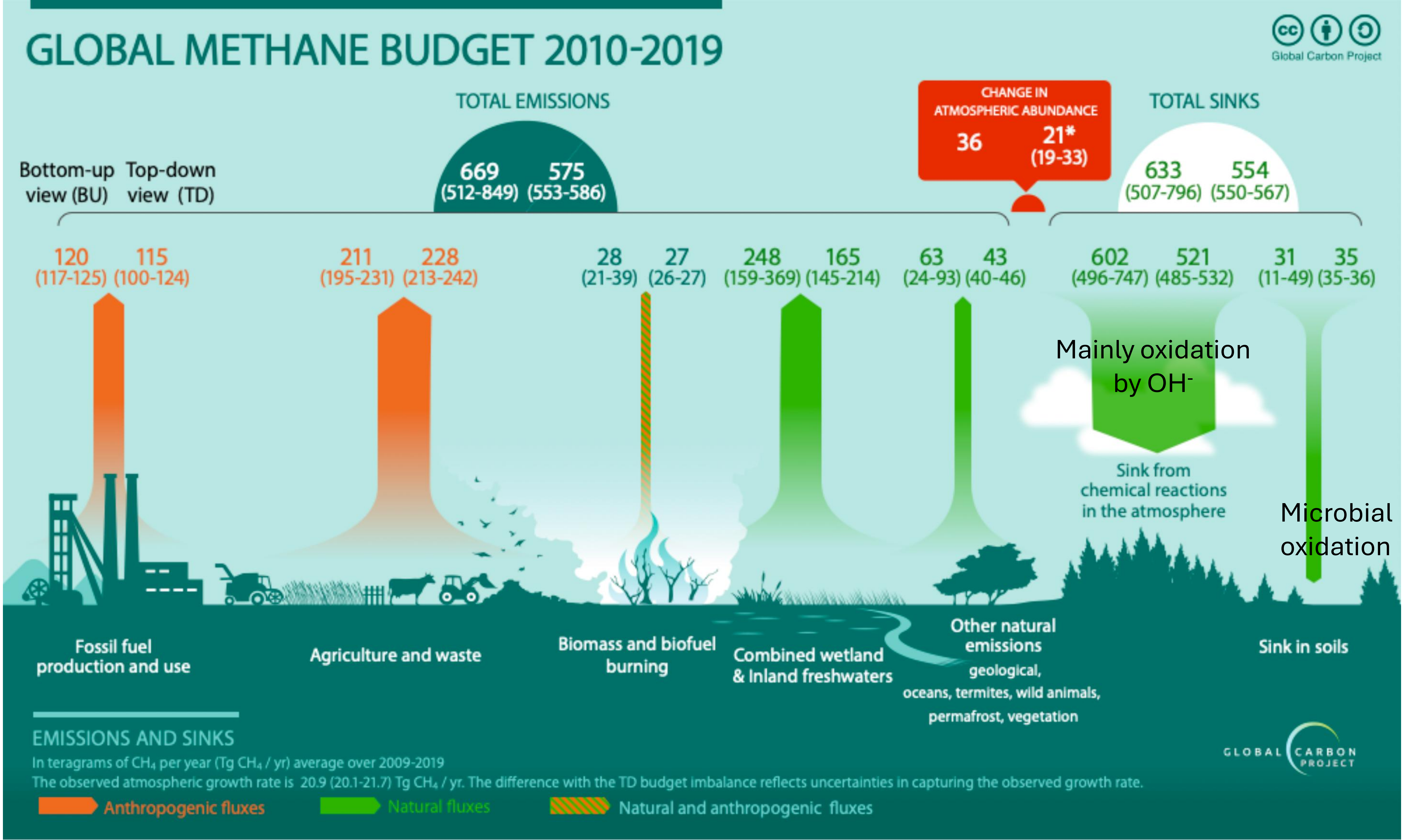
Unlike methanogenesis, methanotrophy exists in both oxygen-free and oxygen-containing environments. Oxygen is used to convert methane, but in environments without oxygen other compounds substitute for it.

FUN FACT
In an example of microbial symbiosis, methanotrophic aggregates of archaea and bacteria convert methane to carbon dioxide.

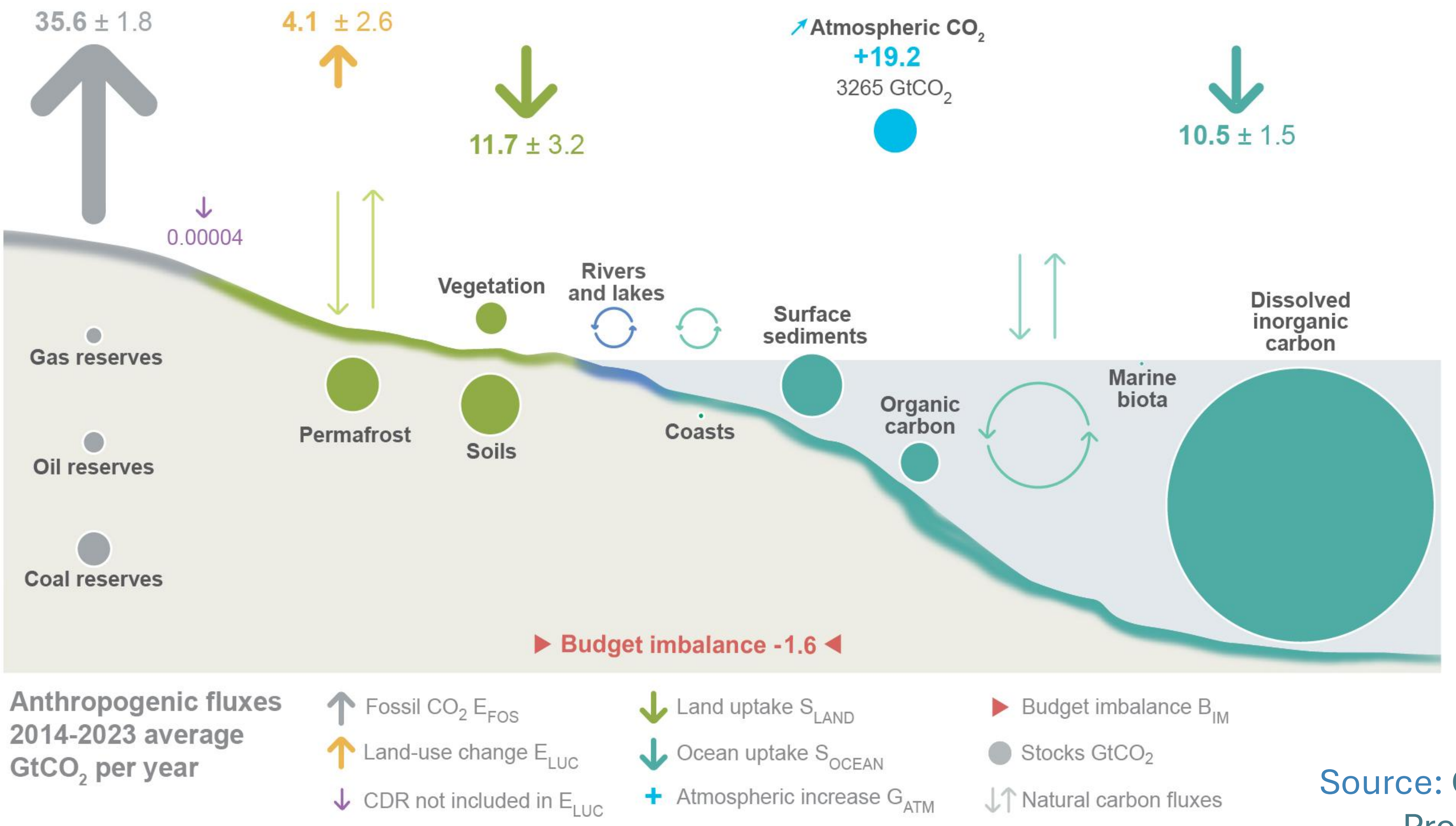
**These are the two most important types of methanotrophy, but many other reactions exist.*

Where in the World is Methane Made?

Where in the World is Methane Munched?


GLOBAL METHANE BUDGET 2010-2019
Global Carbon Project
TOTAL EMISSIONS
669 (512-849) 575 (553-586)
CHANGE IN ATMOSPHERIC ABUNDANCE
36 21* (19-33)
TOTAL SINKS
633 (507-796) 554 (550-567)
Bottom-up view (BU) Top-down view (TD)
120 (117-125) 115 (100-124)
211 (195-231) 228 (213-242)
28 (21-39) 27 (26-27)
248 (159-369) 165 (145-214)
63 (24-93) 43 (40-46)
602 (496-747) 521 (485-532)
31 (11-49) 35 (35-36)
Mainly oxidation by OH-
Sink from chemical reactions in the atmosphere
Microbial oxidation
Fossil fuel production and use
Agriculture and waste
Biomass and biofuel burning
Combined wetland & Inland freshwaters
Other natural emissions
geological, oceans, termites, wild animals, permafrost, vegetation
Sink in soils
EMISSIONS AND SINKS
In teragrams of CH4 per year (Tg CH4 / yr) average over 2009-2019
The observed atmospheric growth rate is 20.9 (20.1-21.7) Tg CH4 / yr. The difference with the TD budget imbalance reflects uncertainties in capturing the observed growth rate.
Anthropogenic fluxes
Natural fluxes
Natural and anthropogenic fluxes
GLOBAL CARBON PROJECT

Anthropogenic perturbation of the global carbon cycle (2014-2023)



Fluxes: $GtCO_2$ per year

The budget imbalance is the difference between the estimated emissions and sinks.

CDR: Carbon Dioxide Removal

Source: Global Carbon Project 2024

Fate of anthropogenic CO_2 emissions (2014–2023)

Sources = Sinks

9.7 GtC/yr

90%

10%

1.1 GtC/yr

5.2 GtC/yr

48%

29%

3.2 GtC/yr

26%

2.9 GtC/yr

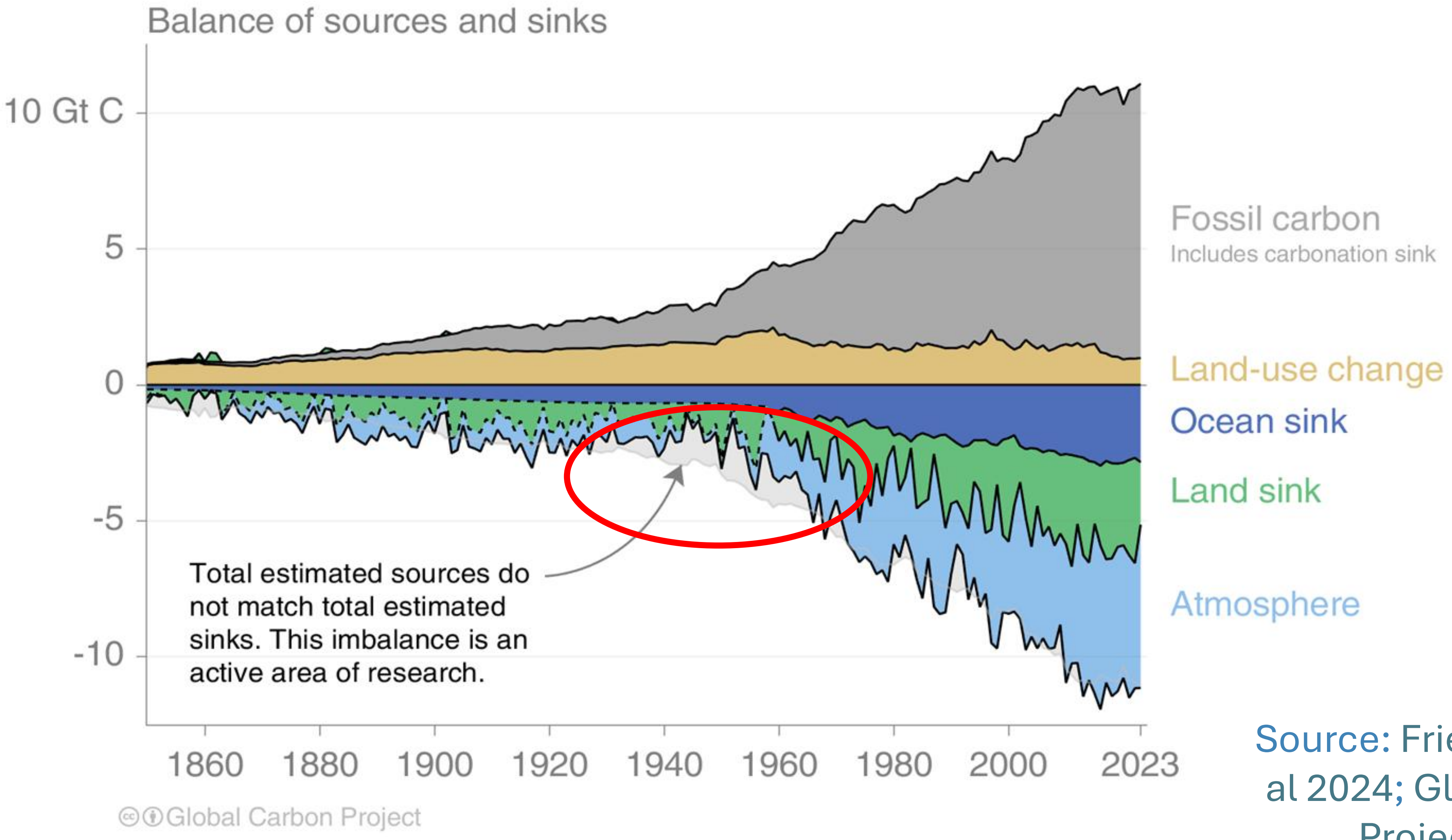
Budget Imbalance:
(the difference between estimated sources & sinks)

4%
-0.4 GtC/yr

Source: Friedlingstein et al 2024; Global Carbon Project 2024

Global anthropogenic carbon budget

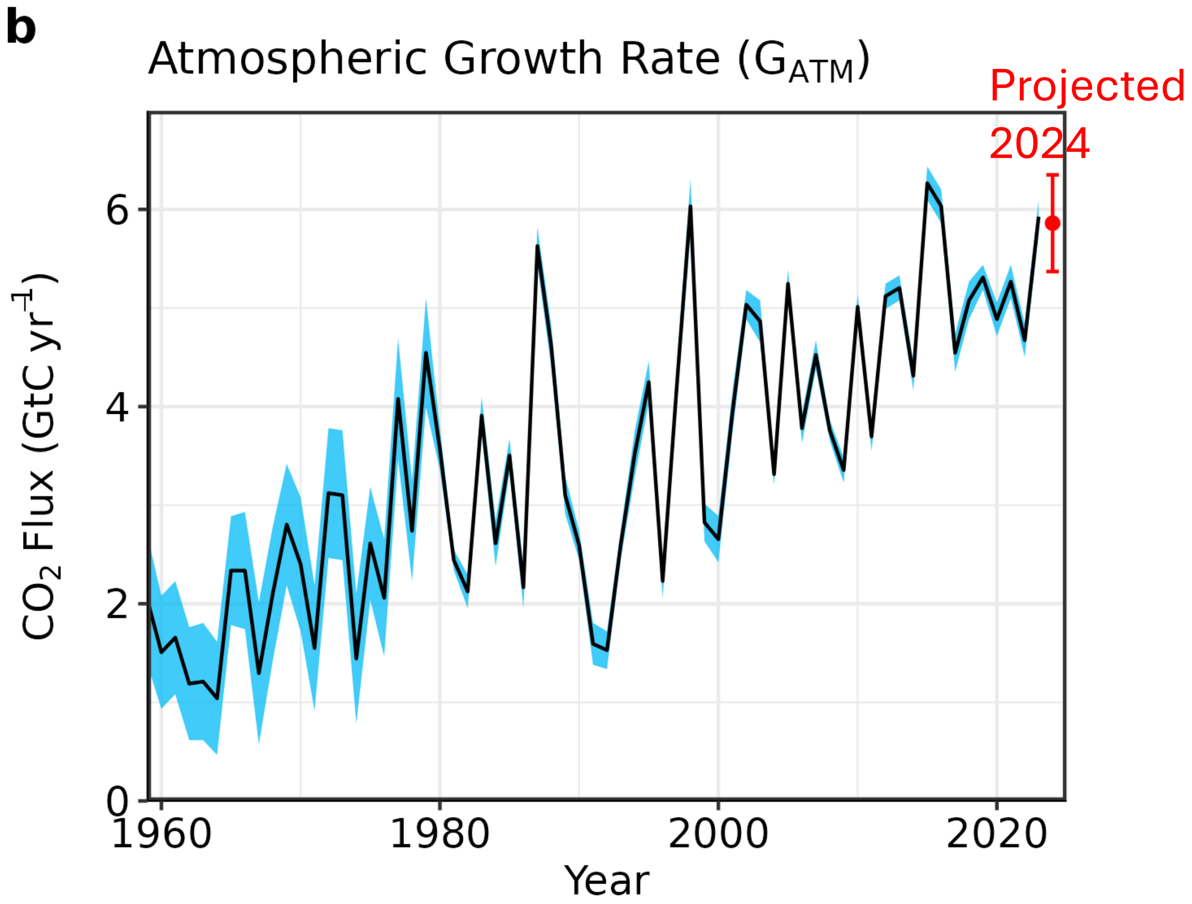
Carbon emissions are partitioned among the atmosphere and carbon sinks on land and in the ocean.



Source: Friedlingstein et al 2024; Global Carbon Project 2024

Atmospheric concentration

The atmospheric concentration growth rate has increased steadily.



Ocean Sink

The ocean carbon sink amounts to 2.9 ± 0.4 GtC/yr for 2014–2023.

Terrestrial sink

The land carbon sink was 3.2 ± 0.9 GtC/yr during 2014–2023.

Source:NOAA-GML; SOCATv2023; Bakker et al 2016; Friedlingstein et al 2024; Global Carbon Project 2024